

Names:	Reg no:
Kommineni Anuradha	AP24110010502
Allada Layachandana	AP24110011636
Chollangi Pragnya sree	AP24110011642
Madireddy Rishitha	AP24110011630

```
CREATE DATABASE IF NOT EXISTS cc;
```

```
USE cc;
```

```
DROP TABLE IF EXISTS Loans;
```

```
DROP TABLE IF EXISTS BookCopies;
```

```
DROP TABLE IF EXISTS Members;
```

```
DROP TABLE IF EXISTS Librarians;
```

```
DROP TABLE IF EXISTS Books;
```

```
CREATE TABLE Books (
```

```
    ISBN VARCHAR(20) NOT NULL,
```

```
    Title VARCHAR(255) NOT NULL,
```

```
    Author VARCHAR(255) NOT NULL,
```

```
    Publisher VARCHAR(150),
```

```
    PubYear SMALLINT,
```

```
    Category VARCHAR(100),
```

```
    CONSTRAINT pk_books PRIMARY KEY (ISBN)
```

```
);
```

```
CREATE TABLE BookCopies (
```

```
    CopyID INT NOT NULL AUTO_INCREMENT,
```

```
ISBN VARCHAR(20) NOT NULL,  
ShelfLocation VARCHAR(50),  
Status VARCHAR(20) NOT NULL DEFAULT 'Available',  
CONSTRAINT pk_bookcopies PRIMARY KEY (CopyID),  
CONSTRAINT fk_bookcopies_book FOREIGN KEY (ISBN)  
REFERENCES Books(ISBN)  
ON UPDATE CASCADE  
ON DELETE CASCADE,  
CONSTRAINT chk_copy_status  
CHECK (Status IN ('Available', 'Borrowed', 'Lost', 'Damaged'))  
);
```

```
CREATE TABLE Members (  
MemberID INT NOT NULL AUTO_INCREMENT,  
FirstName VARCHAR(100) NOT NULL,  
LastName VARCHAR(100) NOT NULL,  
Address VARCHAR(255),  
Phone VARCHAR(20),  
Email VARCHAR(150) UNIQUE,  
MembershipDate DATE NOT NULL DEFAULT (CURRENT_DATE),  
CONSTRAINT pk_members PRIMARY KEY (MemberID)  
);
```

```
CREATE TABLE Librarians (  
LibrarianID INT NOT NULL AUTO_INCREMENT,  
Name VARCHAR(150) NOT NULL,  
Email VARCHAR(150) UNIQUE,  
HireDate DATE NOT NULL,
```

```
CONSTRAINT pk_librarians PRIMARY KEY (LibrarianID)
);
```

```
CREATE TABLE Loans (
    LoanID INT NOT NULL AUTO_INCREMENT,
    CopyID INT NOT NULL,
    MemberID INT NOT NULL,
    LibrarianID INT,
    BorrowDate DATE NOT NULL DEFAULT (CURRENT_DATE),
    DueDate DATE NOT NULL,
    ReturnDate DATE,
    Status VARCHAR(20) NOT NULL DEFAULT 'Borrowed',
    Fine DECIMAL(6,2) NOT NULL DEFAULT 0.00,
```

```
CONSTRAINT pk_loans PRIMARY KEY (LoanID),
```

```
CONSTRAINT fk_loans_copy FOREIGN KEY (CopyID)
    REFERENCES BookCopies(CopyID)
    ON UPDATE CASCADE
    ON DELETE RESTRICT,
```

```
CONSTRAINT fk_loans_member FOREIGN KEY (MemberID)
    REFERENCES Members(MemberID)
    ON UPDATE CASCADE
    ON DELETE RESTRICT,
```

```
CONSTRAINT fk_loans_librarian FOREIGN KEY (LibrarianID)
REFERENCES Librarians(LibrarianID)
ON UPDATE CASCADE
ON DELETE SET NULL,
```

```
CONSTRAINT chk_loan_status
CHECK (Status IN ('Borrowed', 'Returned', 'Overdue')),
```

```
CONSTRAINT chk_return_after_borrow
CHECK (ReturnDate IS NULL OR ReturnDate >= BorrowDate),
```

```
CONSTRAINT chk_due_after_borrow
CHECK (DueDate >= BorrowDate)
```

```
);
```

```
CREATE INDEX idx_loans_member ON Loans(MemberID);
```

```
CREATE INDEX idx_loans_copy ON Loans(CopyID);
```

```
CREATE INDEX idx_loans_status ON Loans(Status);
```

```
CREATE INDEX idx_copies_isbn ON BookCopies(ISBN);
```

```
DELIMITER $$
```

```
CREATE TRIGGER trg_before_loan_insert
BEFORE INSERT ON Loans
FOR EACH ROW
BEGIN
    DECLARE copy_status VARCHAR(20);

    SELECT Status INTO copy_status
    FROM BookCopies
    WHERE CopyID = NEW.CopyID;

    IF copy_status IS NULL THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Invalid CopyID: book copy does not exist.';
    ELSEIF copy_status <> 'Available' THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Book copy is not available for borrowing.';
    END IF;
END$$
```

```
CREATE TRIGGER trg_after_loan_insert
AFTER INSERT ON Loans
FOR EACH ROW
BEGIN
    UPDATE BookCopies
```

```
    SET Status = 'Borrowed'
  WHERE CopyID = NEW.CopyID;
END$$
```

```
CREATE TRIGGER trg_after_loan_update
AFTER UPDATE ON Loans
FOR EACH ROW
BEGIN
  IF NEW.ReturnDate IS NOT NULL
    AND OLD.ReturnDate IS NULL THEN
    UPDATE BookCopies
      SET Status = 'Available'
      WHERE CopyID = NEW.CopyID;
  END IF;
END$$
```

```
DELIMITER ;
```

```
INSERT INTO Books
(ISBN, Title, Author, Publisher, PubYear, Category)
VALUES
('101', 'Fundamentals of Database Systems', 'Elmasri & Navathe', 'Pearson', 2015,
'Computer Science'),
('102', 'Sapiens', 'Yuval Noah Harari', 'Harper', 2015, 'History'),
('103', 'Harry Potter and the Sorcerer"s Stone', 'J.K. Rowling', 'Scholastic', 1998, 'Fiction');
INSERT INTO BookCopies
(ISBN, ShelfLocation, Status)
VALUES
```

```
('101', 'A1-01', 'Available'),  
('101', 'A1-02', 'Available'),  
('102', 'A2-01', 'Available'),  
('103', 'B2-11', 'Available');
```

INSERT INTO Members

(FirstName, LastName, Address, Phone, Email, MembershipDate)

VALUES

```
('Asha', 'Rao', '12 MG Road, Vijayawada', '9000011111', 'asha_rao@gmail.com', '2023-01-15'),  
('Rahul', 'Menon', '45 Park Street, Kochi', '9000022222', 'rahul_menon@gmail.com', '2024-06-01');
```

INSERT INTO Librarians

(Name, Email, HireDate)

VALUES

```
('Priya Nair', 'priya.nair@gmail.com', '2020-04-01'),  
('Sanjay Kumar', 'sanjay.kumar@gmail.com', '2022-09-12');
```

INSERT INTO Loans

(CopyID, MemberID, LibrarianID, BorrowDate, DueDate, Status)

VALUES

```
(1, 1, 1, CURRENT_DATE, DATE_ADD(CURRENT_DATE, INTERVAL 14 DAY), 'Borrowed');
```

INSERT INTO Loans

(CopyID, MemberID, LibrarianID, BorrowDate, DueDate, Status)

VALUES

```
(2, 2, 2, DATE_SUB(CURRENT_DATE, INTERVAL 30 DAY), DATE_SUB(CURRENT_DATE, INTERVAL 16 DAY), 'Borrowed');
```

```
UPDATE Loans
SET Status = 'Overdue'
WHERE Status = 'Borrowed'
AND DueDate < CURRENT_DATE;
```

```
UPDATE Loans
SET ReturnDate = CURRENT_DATE,
    Status = 'Returned'
WHERE LoanID = 1;
```

```
SELECT
    l.LoanID,
    m.FirstName,
    m.LastName,
    b.Title,
    bc.CopyID,
    l.BorrowDate,
    l.DueDate,
    DATEDIFF(CURRENT_DATE, l.DueDate) AS DaysOverdue
FROM Loans l
JOIN BookCopies bc ON l.CopyID = bc.CopyID
JOIN Books b ON bc.ISBN = b.ISBN
JOIN Members m ON l.MemberID = m.MemberID
WHERE l.Status = 'Overdue'
AND l.DueDate < CURRENT_DATE;
```

```
SELECT
```

```
b.ISBN,  
b.Title,  
b.Author,  
COUNT(bc.CopyID) AS AvailableCopies  
FROM Books b  
JOIN BookCopies bc ON b.ISBN = bc.ISBN  
WHERE bc.Status = 'Available'  
GROUP BY b.ISBN, b.Title, b.Author  
HAVING COUNT(bc.CopyID) > 0;
```

```
SELECT  
l.LoanID,  
b.Title,  
bc.CopyID,  
l.BorrowDate,  
l.DueDate,  
l.ReturnDate,  
l.Status,  
l.Fine  
FROM Loans l  
JOIN BookCopies bc ON l.CopyID = bc.CopyID  
JOIN Books b ON bc.ISBN = b.ISBN  
WHERE l.MemberID = 1  
ORDER BY l.BorrowDate DESC;  
SELECT * FROM Books;  
SELECT * FROM BookCopies;  
SELECT * FROM Members;  
SELECT * FROM Librarians;
```

```
SELECT * FROM Loans;
```

	ISBN	Title	Author	Publisher	PubYear	Category
▶	101	Fundamentals of Database Systems	Elmasri & Navathe	Pearson	2015	Computer Science
	102	Sapiens	Yuval Noah Harari	Harper	2015	History
	103	Harry Potter and the Sorcerer's Stone	J.K. Rowling	Scholastic	1998	Fiction
•	NULL	NULL	NULL	NULL	NULL	NULL

	CopyID	ISBN	ShelfLocation	Status
▶	1	101	A1-01	Available
	2	101	A1-02	Borrowed
	3	102	A2-01	Available
	4	103	B2-11	Available
•	NULL	NULL	NULL	NULL

[illegible]

	LibrarianID	Name	Email	HireDate
▶	1	Priya Nair	priya.nair@gmail.com	2020-04-01
	2	Sanjay Kumar	sanjay.kumar@gmail.com	2022-09-12
*	NULL	NULL	NULL	NULL

[illegible]

ER Diagram

LIBRARY LENDING SYSTEM — ER DIAGRAM

